Chapter 1
Laplace Transformation

Definition: For all $t \geq 0$,

$$F(s) = \int_0^\infty e^{-st} f(t)dt.$$

If the resulting integral exists (i.e., has some finite value), it is a function of s, s may be real or complex.

This function $F(s)$ of variable s is called Laplace Transformation of the original function $f(t)$ and will be denoted by $\mathcal{L}\{f(t)\}$, where $\mathcal{L}$ denotes the Laplace transform operator. Thus

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)dt.$$

The original function $f(t)$ is called the inverse transform or inverse of $F(s)$ and will be denoted by $\mathcal{L}^{-1}\{F(s)\}$.

$$\therefore f(t) = \mathcal{L}^{-1}\{F(s)\}.$$

Properties of Laplace transformation:

1. $\mathcal{L}\{af_1(t) + bf_2(t)\} = a\mathcal{L}\{f_1(t)\} + b\mathcal{L}\{f_2(t)\}$ **(linearity)**, where a &b are constants.

2. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$ **(first shifting or translation)**

3. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}[F(s)]$ **(multiplication by t^n)**

Proof of some selective formulae:

Example 1.1

$\mathcal{L}\{1\} = \frac{1}{s}$, $(s > 0)$

Proof: From the definition of Laplace transformation, we know that,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)dt$$

$$\mathcal{L}\{1\} = \int_0^\infty e^{-st}\, 1\, dt$$

$$= \int_0^\infty e^{-st}\, dt$$

$$= \left(\frac{e^{-st}}{-s}\right)\Big|_0^\infty$$

$$= \frac{1}{s} \qquad \text{where } s > 0$$

Example 1.2

$\mathcal{L}\{t\} = \frac{1}{s^2}$, $(s > 0)$

Proof: From the definition of Laplace transformation, we know that,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)dt$$

$$\begin{aligned}
\mathcal{L}\{t\} &= \int_0^\infty e^{-st}\, t\, dt \\
&= t\left(\frac{e^{-st}}{-s}\right)\Big|_0^\infty + \int_0^\infty \frac{e^{-st}}{s}\, dt \\
&= 0 + \frac{1}{s}\int_0^\infty e^{-st}\, dt \\
&= \frac{e^{-st}}{-s^2}\Big|_0^\infty \\
&= \frac{1}{s^2} \qquad \text{where } s > 0
\end{aligned}$$

Example 1.3

$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, (s > a)$

Proof: From the definition of Laplace transformation, we know that,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

$$\begin{aligned}
\mathcal{L}\{e^{at}\} &= \int_0^\infty e^{-st} e^{at} dt \\
&= \lim_{p\to\infty} \int_0^p e^{-(s-a)t} dt \\
&= \lim_{p\to\infty} \frac{e^{-(s-a)t}}{-(s-a)}\Big|_0^p \\
&= \lim_{p\to\infty} \frac{1 - e^{-(s-a)p}}{(s-a)} \\
&= \frac{1}{s-a} \qquad \text{where } s > a
\end{aligned}$$

Example 1.4

$\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$, if $|s| > a$ and $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$, if $|s| > a$

Proof:

$$\begin{aligned}
\mathcal{L}\{e^{iat}\} &= \frac{1}{s-ia} \\
&= \frac{s+ia}{(s-ia)(s+ia)} \\
&= \frac{s+ia}{s^2+a^2} \\
&= \frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}
\end{aligned}$$

But $e^{iat} = \cos at + i \sin at$

So,

$$\mathcal{L}\{e^{iat}\} = \int_0^\infty e^{-st}e^{iat}dt$$
$$= \int_0^\infty e^{-st}(\cos at + i\sin at)dt$$
$$= \int_0^\infty e^{-st}\cos at\, dt + i\int_0^\infty e^{-st}\sin at\, dt$$
$$= \mathcal{L}\{\cos at\} + i\,\mathcal{L}\{\sin at\}$$

Comparing (1) and (2), we have on equating real and imaginary parts,
$\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$ and $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$ if $|s| > a$.

Some related important formulae	
$\sin at = \frac{e^{iat} - e^{-iat}}{2i}$	$\cos at = \frac{e^{iat} + e^{-iat}}{2}$
$\sinh at = \frac{e^{at} - e^{-at}}{2}$	$\cosh at = \frac{e^{at} + e^{-at}}{2}$
$\int e^{at}\sin bt\ dt = \frac{e^{at}(a\sin bt - b\cos bt)}{a^2+b^2}$	$\int e^{at}\cos bt\ dt = \frac{e^{at}(a\cos bt + b\sin bt)}{a^2+b^2}$

Table of Some Important Formulae Related to Laplace Transformation:

1.	$\mathcal{L}\{1\} = \frac{1}{s}, \quad (s>0)$	2.	$\mathcal{L}\{c\} = \frac{c}{s}$, c is any constant, $(s>0)$
3.	$\mathcal{L}\{t\} = \frac{1}{s^2}, \quad (s>0)$	4.	$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$, $n = 0,1,2,3,\ldots$ and $s>0$
5.	$\mathcal{L}\{e^t\} = \frac{1}{s-1}, \quad (s>1)$	6.	$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad (s>a)$
7.	$\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}, \quad (s>0)$	8.	$\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}, (s>0)$
9.	$\mathcal{L}\{\sinh at\} = \frac{a}{s^2-a^2}, \lvert s\rvert > a$	10.	$\mathcal{L}\{\cosh at\} = \frac{s}{s^2-a^2}, \lvert s\rvert > a$
11.	$\mathcal{L}\{e^{at}\sin\omega t\} = \frac{\omega}{(s-a)^2+\omega^2}$ $(s>a)$	12.	$\mathcal{L}\{e^{at}\cos\omega t\} = \frac{s-a}{(s-a)^2+\omega^2}$ $(s>a)$
13.	$\mathcal{L}\{e^{at}\sinh\omega t\} = \frac{\omega}{(s-a)^2-\omega^2}$ $(\lvert s-a\rvert > \omega)$	14.	$\mathcal{L}\{e^{at}\cosh\omega t\} = \frac{s-a}{(s-a)^2-\omega^2}$ $(\lvert s-a\rvert > \omega)$
15.	$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}[F(s)]$		

Some workout examples on Laplace transformation:

Example 1.5

$$\begin{aligned}\mathcal{L}\{(t^2+1)^2\} &= \mathcal{L}\{t^4+2t^2+1\}\\ &= \mathcal{L}\{t^4\}+2\mathcal{L}\{t^2\}+\mathcal{L}\{1\}\\ &= \frac{4!}{s^{4+1}}+2\times\frac{2!}{s^{2+1}}+\frac{1}{s}\\ &= \frac{24}{s^5}+\frac{4}{s^3}+\frac{1}{s}\end{aligned}$$

$$\begin{aligned}\mathcal{L}\{\mathrm{e}^{-3t}+5\cosh t\} &= \mathcal{L}\{\mathrm{e}^{-3t}\}+5\,\mathcal{L}\{\cosh\ t\}\\ &= \frac{1}{\mathrm{s}-(-3)}+5\frac{\mathrm{s}}{\mathrm{s}^2-1^2}\\ &= \frac{1}{\mathrm{s}+3}+\frac{5\mathrm{s}}{\mathrm{s}^2-1}\end{aligned}$$

Example 1.6

$$\begin{aligned}\mathcal{L}\{(\sin t-\cos t)^2+cos^2\,3t\} &= \mathcal{L}\left\{\sin^2 t-2\sin t\cos t+\cos^2 t+\frac{1}{2}(1+\cos 6t)\right\}\\ &= \mathcal{L}\left\{1-\sin 2t+\frac{1}{2}+\frac{1}{2}\cos 6t)\right\}\\ &= \mathcal{L}\left\{\frac{3}{2}\right\}-\mathcal{L}\{\sin 2t\}+\frac{1}{2}\mathcal{L}\{\cos 6t\}\\ &= \frac{3}{2s}-\frac{2}{s^2+2^2}+\frac{1}{2}\frac{s}{s^2+6^2}\\ &= \frac{3}{2s}-\frac{2}{s^2+4}+\frac{1}{2}\left(\frac{s}{s^2+36}\right)\end{aligned}$$

Example 1.7

$$\begin{aligned}\mathcal{L}\{3-\mathrm{e}^{-t}\cos 2t+t\ \sinh 3t\} &= \mathcal{L}\{3\}-\mathcal{L}\{\mathrm{e}^{-t}\cos 2t\}+\mathcal{L}\{t\sinh 3t\}\\ &= \frac{3}{\mathrm{s}}-\frac{\mathrm{s}+1}{(\mathrm{s}+1)^2+2^2}+(-1)^1\frac{\mathrm{d}}{\mathrm{ds}}\left(\frac{3}{\mathrm{s}^2-3^2}\right)\\ &= \frac{3}{\mathrm{s}}-\frac{\mathrm{s}+1}{\mathrm{s}^2+2\mathrm{s}+5}+3\times\frac{2\mathrm{s}}{(\mathrm{s}^2-9)^2}\\ &= \frac{3}{\mathrm{s}}-\frac{\mathrm{s}+1}{\mathrm{s}^2+2\mathrm{s}+5}+\frac{6\mathrm{s}}{(\mathrm{s}^2-9)^2}\end{aligned}$$

Example 1.8

$$\begin{aligned}\mathcal{L}\{t\cos t-\mathrm{e}^{-3t}\sin 2t+t^7\mathrm{e}^{5t}\} &= (-1)^1\frac{\mathrm{d}}{\mathrm{ds}}\left(\frac{\mathrm{s}}{\mathrm{s}^2+1}\right)-\frac{2}{(\mathrm{s}+3)^2+2^2}+\frac{7!}{(\mathrm{s}-5)^8}\\ &= -\frac{1-\mathrm{s}^2}{(\mathrm{s}^2+1)^2}-\frac{2}{(\mathrm{s}+3)^2+4}+\frac{5040}{(\mathrm{s}-5)^8}\end{aligned}$$

Example 1.9

$$\begin{aligned}\mathcal{L}\{\mathrm{e}^{2t}t+\mathrm{e}^t\cosh 3t\} &= \frac{1}{(\mathrm{s}-2)^2}+\frac{\mathrm{s}-1}{(\mathrm{s}-1)^2-3^2}\\ &= \frac{1}{(\mathrm{s}-2)^2}+\frac{\mathrm{s}-1}{\mathrm{s}^2-2\mathrm{s}-8}\end{aligned}$$

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Example 1.10

$$\begin{aligned}
\mathcal{L}\{t\sin at\} &= \mathcal{L}\left\{t\frac{e^{iat}-e^{-iat}}{2i}\right\}\\
&= \frac{1}{2i}\mathcal{L}\{te^{iat}-te^{-iat}\}\\
&= \frac{1}{2i}\left[\frac{1!}{(s-ia)^2}-\frac{1!}{(s+ia)^2}\right] \quad \text{(first Shifting Property)}\\
&= \frac{2as}{(s^2+a^2)^2}
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\mathcal{L}\{t\sin at\} &= (-1)^1\frac{d}{ds}\left(\frac{a}{s^2+a^2}\right)\\
&= -\frac{(s^2+a^2).0-a.2s}{(s^2+a^2)^2}\\
&= -\frac{-2as}{(s^2+a^2)^2}\\
&= \frac{2as}{(s^2+a^2)^2}
\end{aligned}$$

Example 1.11

$$\begin{aligned}
\mathcal{L}\{t\cos at\} &= \mathcal{L}\left\{t\frac{e^{iat}+e^{-iat}}{2}\right\}\\
&= \frac{1}{2}\mathcal{L}\{e^{iat}t+e^{-iat}t\}\\
&= \frac{1!}{2(s-ia)^2}+\frac{1!}{2(s+ia)^2} \quad \text{(first shifting property)}\\
&= \frac{(s+ia)^2+(s-ia)^2}{2(s-ia)^2(s+ia)^2}\\
&= \frac{(s^2+2ias+i^2a^2)+(s^2-2ias+i^2a^2)}{2\{(s-ia)(s+ia)\}^2}\\
&= \frac{2(s^2-a^2)}{2(s^2-i^2a^2)^2}\\
&= \frac{s^2-a^2}{(s^2+a^2)^2}
\end{aligned}$$

Alternatively

$$\begin{aligned}
\mathcal{L}\{t\cos at\} &= (-1)^1\frac{d}{ds}\left(\frac{s}{s^2+a^2}\right)\\
&= -\frac{(s^2+a^2).1-s.2s}{(s^2+a^2)^2}\\
&= \frac{s^2-a^2}{(s^2+a^2)^2}
\end{aligned}$$

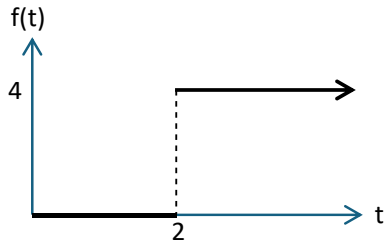
Piecewise function

Example 1.12

Sketch $f(t) = \begin{cases} 0, & 0 < t < 2 \\ 4, & t > 2 \end{cases}$ and

Find $\mathcal{L}\{f(t)\}$.

Solution:



$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)dt$$

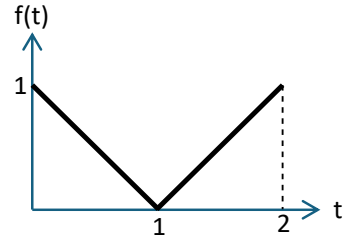
$$= \int_0^2 e^{-st}.0\ dt + \int_2^\infty e^{-st}\ 4\ dt$$

$$= 4\left[\frac{e^{-st}}{-s}\right]\Bigg|_2^\infty$$

$$= \frac{4e^{-2s}}{s}$$

Example 1.13

Define the function $f(t)$ and find the Laplace transformation of $f(t)$ by using definition



Solution: The function $f(t)$ is defined as

$$f(t) = \begin{cases} 1-t, & 0 \le t \le 1 \\ t-1, & 1 \le t < 2 \\ 0, & t \ge 2 \end{cases}$$

By the definition of Laplace transformation,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)dt$$

$$= \int_0^1 e^{-st} f(t)dt + \int_1^2 e^{-st} f(t)dt + \int_2^\infty e^{-st} f(t)dt$$

$$= \int_0^1 e^{-st}(1-t)dt + \int_1^2 e^{-st}(t-1)dt + \int_2^\infty e^{-st}.0dt$$

$$= \int_0^1 e^{-st}dt - \int_0^1 t\, e^{-st}dt + \int_1^2 t\, e^{-st}dt - \int_1^2 e^{-st}dt$$

$$= \left[\frac{e^{-st}}{-s}\right]\Big|_0^1 - \left[\frac{te^{-st}}{-s} - \frac{e^{-st}}{s^2}\right]\Big|_0^1 + \left[\frac{te^{-st}}{-s} - \frac{e^{-st}}{s^2}\right]\Big|_1^2 - \left[\frac{e^{-st}}{-s}\right]\Big|_1^2$$

$$= \frac{1}{s} - \frac{1}{s^2} + \frac{2e^{-s}}{s^2} - \frac{e^{-2s}}{s} - \frac{e^{-2s}}{s^2}.$$

Reference Books:

1. Advanced Engineering Mathematics- Erwin Kreyszig. (10th Edition)
2. Differential Equations- Paul Blanchard, Robert L. Devaney, Glen R. Hall (4th Edition)

Problem Set 1.1

1. **Find the Laplace Transforms and sketch (if free hand sketching is getting complex then use plotting software) the following functions:**

i. <u>Using direct formula</u>

a. $f(t) = e^{5t}$, **Ans:** $F(s) = \frac{1}{s-5}$.

b. $f(t) = e^{-2t}$, **Ans:** $F(s) = \frac{1}{s+2}$.

c. $f(t) = sin(\omega t + \theta)$, **Ans:** $F(s) = \frac{\omega\, cos\,\theta + s\ sin\,\theta}{\omega^2 + s^2}$.

d. $f(t) = 1.5 \sin\left(3t - \frac{\pi}{2}\right)$, **Ans:** $F(s) = \frac{-3s}{2(s^2+9)}$.

ii. <u>First shifting or Translation property</u>

If $\mathcal{L}\{f(t)\} = F(s)$ then $\mathcal{L}\{e^{at}\, f(t)\} = F(s-a)$.

e. $f(t) = e^{2t} \sinh t$, **Ans:** $F(s) = \frac{1}{(s-2)^2-1}$.

f. $f(t) = e^{-t} \sinh 4t$, **Ans:** $F(s) = \frac{4}{(s+1)^2-16}$.

g. $f(t) = e^{2t} \cos 3t$, **Ans:** $F(s) = \frac{s-2}{(s-2)^2\ +\ 9}$.

h. $f(t) = t^{10}\, e^{-7t}$, **Ans:** $F(s) = \frac{10\,!}{(s+7)^{11}}$.

i. $f(t) = \cosh 5t\ \ e^{3t}$, **Ans:** $F(s) = \frac{s-3}{(s-3)^2\ -25}$.

iii. <u>Property of multiplication by t^n</u>

If $\mathcal{L}\{f(t)\} = F(s)$ then $\mathcal{L}\{t^n\, f(t)\} = (-1)^n\ \frac{d^n}{ds^n}[F(s)]$.

j. $f(t) = t \sin 2t$, **Ans:** $F(s) = \frac{4s}{(s^2+4)^2}$.

k. $f(t) = t \cos 3t$, **Ans:** $F(s) = \frac{s^2-9}{(s^2+9)^2}$.

iv. <u>Piece-wise Function</u>

If $f(t)$ is a piece-wise function, where $f(t) = \begin{cases} f_1(t); 0 < t < a \\ f_2(t); \quad t > a \end{cases}$ then

$$\mathcal{L}\{f(t)\} = \int_0^a e^{-st} f_1(t)\ dt + \int_a^\infty e^{-st}\ f_2(t)\ dt$$

l. $f(t) = \begin{cases} 1-t\ ; 0 < t < 1 \\ 0\ ; \quad t > 1 \end{cases}$
Ans: $F(s) = \frac{s+e^{-s}-1}{s^2}$.

m. $f(t) = \begin{cases} 1\ ; \ 0 < t < 2 \\ 0\ ; \quad t > 2 \end{cases}$
Ans: $F(s) = \frac{1}{s} - \frac{e^{-2s}}{s}$

n. $f(t) = \begin{cases} 3-t\ ; 0 < t < 3 \\ 0\ ; \quad t > 3 \end{cases}$
Ans: $F(s) = -\frac{1}{s^2} + \frac{e^{-3s}}{s^2} + \frac{3}{s}$.

o. $f(t) = \begin{cases} 1\ ; 0 < t < 1 \\ -1\ ; 1 < t < 2 \\ 0; \quad t > 2 \end{cases}$
Ans: $F(s) = \frac{1}{s} + \frac{e^{-2s}}{s} - \frac{2e^{-s}}{s}$.

p. $f(t) = \begin{cases} t\ ; 0 \le t \le 4 \\ 1\ ; \quad t > 4 \end{cases}$
Also sketch $f(t)$.
Ans: $F(s) = \frac{1}{s^2} - \frac{e^{-4s}}{s^2} - \frac{3e^{-4s}}{s}$.

2. **Problems from Book: (**Advanced Engineering Mathematics – Erwin Kreyszig**)**

i. Find the Laplace transformation using definition:

a. $\mathcal{L}\{e^{at}\}$
b. $\mathcal{L}\{t^2 e^{at}\}$
c. $\mathcal{L}\{sin\, 2\, t\}$
d. $\mathcal{L}\{cos\, a\, t\}$
e. $\mathcal{L}\{cos\, 3\, t\}$

ii. Find the Laplace transformation of the following functions, assuming *a, b, ω, π* as constants:

a. $f(t) = 3t + 12$
Ans: $\frac{3}{s^2} + \frac{12}{s}$

b. $f(t) = e^{2t} \sinh t$
Ans: $\frac{1}{(s-2)^2-1}$

c. $f(t) = (5-6t)^2$
Ans: $\frac{72}{s^3} - \frac{60}{s^2} + \frac{25}{s}$

d. $f(t) = e^{-t} \sinh 4\, t$
Ans: $\frac{4}{(s+1)^2-16}$

e. $f(t) = \cos^2 \omega t$
Ans: $\frac{1}{2s} + \frac{s}{2(s^2+4\omega^2)}$

f. $f(t) = t^5 e^{2t}$
Ans: $\frac{5!}{(s-2)^6}$

g. $f(t) = \cos \pi\, t$
Ans: $\frac{s}{s^2+\pi^2}$

h. $f(t) = \sin^2\left(\frac{1}{2}\pi t\right)$
Ans: $\frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+\pi^2}\right]$

i. $f(t) = \frac{1+\cos(2\omega t)}{2}$

j. $f(t) = 12te^{-3t}$

Ans: $\frac{1}{2}\left[\frac{1}{s}+\frac{s}{s^2+4\omega^2}\right]$

k. $f(t)=5\cosh 2t-3\sinh t$

Ans: $\frac{5s}{s^2-4}-\frac{3}{s^2-1}$

l. $f(t)=t^2e^{-3t}$

Ans: $\frac{12}{(s+3)^2}$

m. $f(t)=e^{-t}(\cos 4t-2\sin 4t)$

Ans: $\frac{s-7}{(s+1)^2+16}$

n. $f(t)=t\cos t+\sin t$

Ans: $\frac{2}{(s+3)^3}$

o. $\boldsymbol{f(t)=2e^{-3t}cos\,4t}$

Ans: $\frac{s^2-1}{(s^2+1)^2}+\frac{1}{s^2+1}$

p. $f(t)=e^t(t^2+\sin 3t)$

Ans: $\frac{2(s+3)}{s^2+6s+25}$

q. $\boldsymbol{f(t)=3e^{-4t}sin\,5\pi t}$

Ans: $\frac{15\pi}{(s+4)^2+25\pi^2}$

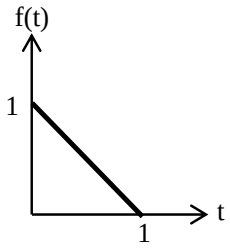
r. $f(t)=e^{-3t}(t+4)^2$

Ans: $\frac{2}{(s-1)^3}+\frac{3}{(s-1)^2+9}$

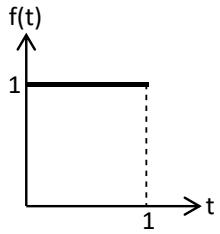
Ans: $\frac{2}{(s+3)^3}+\frac{8}{(s+3)^2}+\frac{16}{s+3}$

iii. Define the function $f(t)$ and find the Laplace transformation of $f(t)$ by using definition

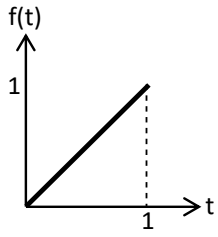
a.



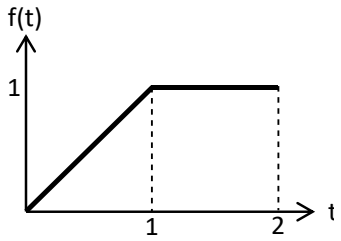
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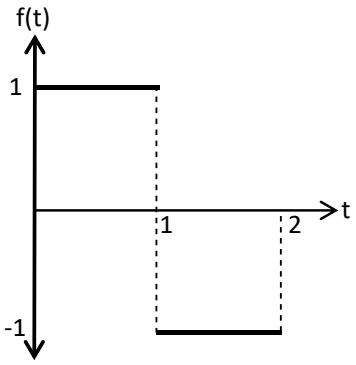
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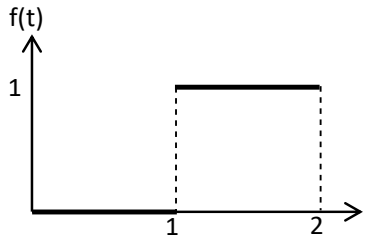
d.



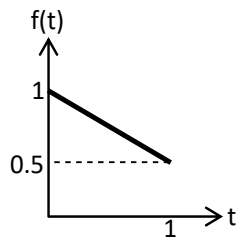
e.



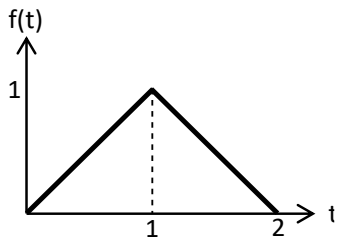
f.



g.



h.



The Unit step function (Heaviside function)

In engineering applications, we frequently encounter functions whose values change abruptly at specified values of time *t*. One common example is when a voltage is switched on or off in an electrical circuit at a specified value of time *t*.

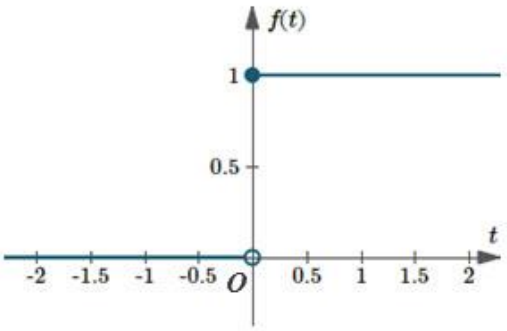
The value of $t = 0$ is usually taken as a convenient time to switch on or off the given voltage.

The switching process can be described mathematically by the function called the **Unit Step Function** (otherwise known as the **Heaviside function** after Oliver Heaviside).

The Unit step function or Heaviside's unit step function, $u(t)$,is defined as follows:

$$f(t) = u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases},$$

That is, $u$is a function of time *t*, and *u* has value **zero** when time is negative (before we flip the switch); and value **one** when time is positive or zero (from when we flip the switch).



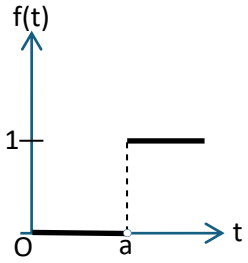
Graph of $f(t) = u(t)$, the unit step function.

Shifted (Right) Unit step function:

In many circuits, waveforms are applied at specified intervals starting at a different initial point other than t=0. Such a function may be described using the **shifted** (aka **delayed**) unit step function.

A function which has value 0 up to the time $t = a$ and thereafter has value 1 is written:

$$u_a(t) = u(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases},$$



Rectangular pulse:

A common situation in a circuit is for a voltage, $V(t)$ to be applied at a particular time (say $t = a$) and removed later, at $t = b$ (say). We write such a situation using unit step functions as:

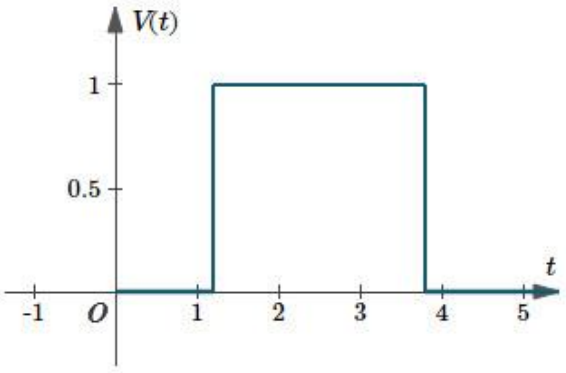
$$V(t) = u(t-a) - u(t-b)$$

This voltage has strength 1, duration $(b-a)$.

Alternatively, $V(t)$ may be constructed using top hat function $u(t-a) - u(t-b)$ as follows:

$$V(t) = u(t-a) - u(t-b) = \begin{cases} 1, & a \leq t < b \\ 0, & \text{otherwise'} \end{cases}$$

The graph of $V(t) = u(t-1.2) - u(t-3.8)$ is as follows. Here, the duration is 3.8−1.2=2.6 .



Graph of $V(t) = u(t-1.2) - u(t-3.8)$ is an example of a rectangular pulse.

Example 1.14

Write the following function in terms of unit step function(s):

(i) $f(t) = 1$, when $t \geq 2$
Ans: $f(t) = 1\, u(t-2)$

(ii) $f(t) = 2$, when $0 \leq t \leq 3$
Ans: $f(t) = 2\{u(t) - u(t-3)\} = 2\{1 - u(t-3)\}$

(iii) $f(t) = t$, when $t \geq 1$
Ans: $f(t) = t\, u(t-1)$

(iv) $f(t) = t^2$, when $t \leq 4$
Ans: $f(t) = t^2\{u(t) - u(t-4)\} = t^2\{1 - u(t-4)\}$

(v) $f(t) = e^t$, when $2 \leq t \leq 3$
Ans: $f(t) = e^t\{u(t-2) - u(t-3)\}$

(vi) $f(t) = e^{t-2}$, when $2 \leq t < \infty$
Ans: = $e^{t-2}\, u(t-2)$

(vii) $$f(t) = \begin{cases} 4, & 0 \leq t < 1 \\ t+3, & 1 \leq t < 2 \\ t^2+1, & 2 \leq t < 3 \\ 10e^{t-3}, & t > 3 \end{cases}$$
Ans: $4\{1 - u(t-1)\} + (t+3)\{u(t-1) - u(t-2)\} + (t^2+1)\{u(t-2) - u(t-3)\} + 10e^{t-3}u(t-3)$

Example 1.15

Write the following function in terms of unit step function(s):

(i) $f(t) = 1$, when $t \geq 2$
Find $\mathcal{L}\{f(t)\}$.
$\mathcal{L}\{u(t-1)\} = \dfrac{e^{-s}}{s}$
Ans: $\dfrac{e^{-s}}{s}$

(ii) $f(t) = 2$, when $0 \leq t \leq 3$
Ans: $f(t) = 2\{u(t) - u(t-3)\} = 2\{1 - u(t-3)\}$

(iii) $f(t) = t$, when $t \geq 1$
Ans: $f(t) = t\, u(t-1)$

(iv) $f(t) = t^2$, when $t \leq 4$
Ans: $f(t) = t^2\{u(t) - u(t-4)\} = t^2\{1 - u(t-4)\}$

(v) $f(t) = e^t$, when $2 \leq t \leq 3$
Ans: $f(t) = e^t\{u(t-2) - u(t-3)\}$

(vi) $f(t) = e^{t-2}$, when $2 \leq t < \infty$
Ans: = $e^{t-2}\, u(t-2)$

Example 1.16

Write the following functions in terms of **unit step** function(s). Sketch each waveform.

(i) A 12-V source is switched on at t = 4 s,

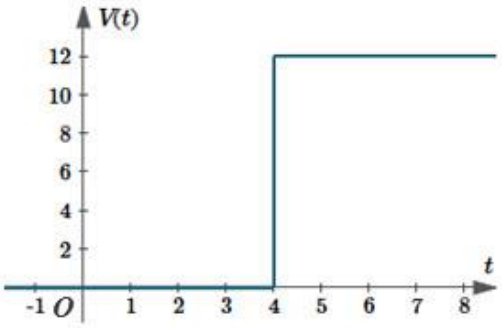
(ii) $$V(t) = \begin{cases} 1, & 0 < t < a \\ 0, & t > a \end{cases}$$
and,

(iii) $$V(t) = \begin{cases} 0, & t < 3 \\ 2t+8, & 3 < t < 5 \\ 0, & t > 5 \end{cases}.$$

Solution:

(i) Since the voltage is turned on at $t = 4$, we need to use $u(t-4)$.We multiply by 12 since that is the voltage.

We write the function as follows: $V(t) = 12 \cdot u(t-4)$
Here's the graph:



Graph of $V(t) = 12 \cdot u(t-4)$

(ii) $V(t) = \begin{cases} 1, & 0 < t < a \\ 0, & t > a \end{cases}$

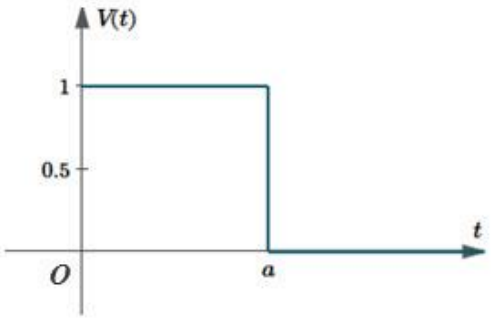
In words, the voltage has value 1 until time $t = a$. Then it is turned off.
We have a "rectangular pulse" situation and need to use this formula:

$$V(t) = u(t-a) - u(t-b)$$

In our example, the pulse starts at $t = 0$, so we use $u(t)$and finishes at $t = a$ so we use $u(t-a)$. So, the required function is:

$$V(t) = 1 \cdot [u(t) - u(t-a)]$$

Here is the graph



Graph of $V(t) = 1 \cdot [u(t) - u(t-a)]$, a shifted unit step function

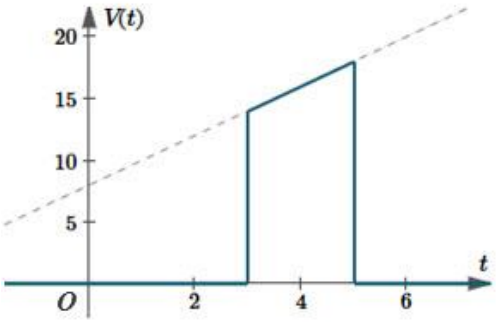
(iv) $$V(t) = \begin{cases} 0, & 0 < t < 3 \\ 2t + 8, & 3 < t < 5 \\ 0, & t > 5 \end{cases}$$

In this example, our function is $V(t) = 2t + 8$ which has slope 2 and V-intercept 8.
The signal is only turned on between $t = 3$ and $t = 5$. The rest of the time, it is off.

So, our voltage function will be:

$$V(t) = (2t + 8) \cdot [u(t - 3) - u(t - 5)]$$

The graph is as follows:



Graph of $V(t) = (2t + 8) \cdot [u(t - 3) - u(t - 5)]$. The dashed line is $V(t) = (2t + 8)$.

Now, the Laplace transform of **unit step function** is,

$$\mathcal{L}\{1.u(t - a)\} = \frac{e^{-as}}{s}$$

$$\mathcal{L}\{f(t).u(t - a)\} = e^{-as}\mathcal{L}\{f(t + a)\}$$

Example 1.17

Find the Laplace transformation of $t^2u(t - 3)$.

Solution:

$$\begin{aligned}\mathcal{L}\{t^2u(t - 3)\} &= e^{-3s}\mathcal{L}\{(t + 3)^2\} \\ &= e^{-3s}\mathcal{L}\{t^2 + 6t + 9\} \\ &= e^{-3s}\left(\frac{2}{s^3} + \frac{6}{s^2} + \frac{9}{s}\right)\end{aligned}$$

Example 1.18

Find the Laplace transformation of $e^{-2t}u(t - \pi)$.

Solution: Here $f(t) = e^{-2t}$ and $u(t - \pi) = \begin{cases}0, & t < \pi \\ 1, & t > \pi\end{cases}$

$$\begin{aligned}\mathcal{L}\{e^{-2t}u(t - \pi)\} &= e^{-\pi s}\mathcal{L}\{f(t + \pi)\} \\ &= e^{-\pi s}\mathcal{L}\{e^{-2(t+\pi)}\} \\ &= e^{-\pi s}\mathcal{L}\{e^{-2t}e^{-2\pi}\} \\ &= e^{-\pi s}e^{-2\pi}\mathcal{L}\{e^{-2t}\} \\ &= e^{-\pi(s+2)}\frac{1}{s + 2}.\end{aligned}$$

Example 1.19

Find the Laplace transformation of $tu(t - 2)$.

Solution:

$$\begin{aligned}\mathcal{L}\{tu(t - 2)\} &= e^{-2s}\mathcal{L}\{t + 2\} \\ &= e^{-2s}\left(\frac{1}{s^2} + 2\frac{1}{s}\right) \\ &= e^{-2s}\left(\frac{2}{s} + \frac{1}{s^2}\right)\end{aligned}$$

Example 1.20

Given $f(t) = \begin{cases} 8, & 0 < t < 2 \\ 6, & t > 2 \end{cases}$,

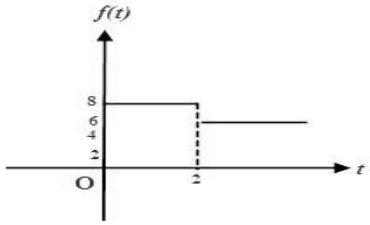
(i) sketch $f(t)$,

(ii) convert $f(t)$to unit step function and,

(iii) find the Laplace transformation of$f(t)$.

Solution:

(i)



(ii) $f(t) = 8[u(t-0) - u(t-2)] + 6[u(t-2)]$
$= 8[u(t-0)] - 2[u(t-2)]$

(iii) $\mathcal{L}\{f(t)\} = \mathcal{L}\{8\} - 2\mathcal{L}\{u(t-2)\}$
$= \frac{8}{s} - \frac{2e^{-2s}}{s}$.

Example 1.21

Given $f(t) = \begin{cases} E, & a < t < b \\ 0, & t > b \end{cases}$, where E, *a* and *b* are positive constants.

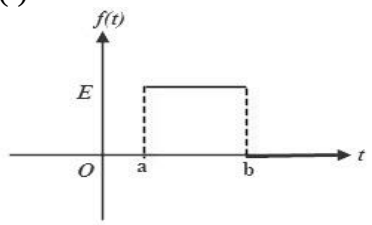
(i) sketch $f(t)$,

(ii) convert $f(t)$to unit step function and,

(iii) find the Laplace transformation of$f(t)$.

Solution:

(i)



(ii) $f(t) = E\begin{cases} 1 & a < t < b \\ 0 & t > b \end{cases}$
$= E[u(t-a) - u(t-b)]$

(iii) $\mathcal{L}\{f(t)\} = E\mathcal{L}\{u(t-a) - u(t-b)\}$
$= E\left[\frac{e^{-as}}{s} - \frac{e^{-bs}}{s}\right]$.

Example 1.22

Given$f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3 \end{cases}$,

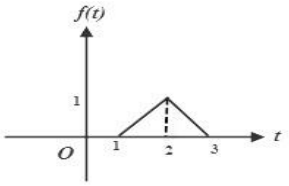
(i) sketch $f(t)$,

(ii) convert $f(t)$to unit step function and,

(iii) find the Laplace transformation of$f(t)$.

Solution:

(i)



(ii) $f(t) = (t-1)[u(t-1) - u(t-2)]$
$+ (3-t)[u(t-2) - u(t-3)]$
$= (t-1)u(t-1) - (t-1)u(t-2)$
$+ (3-t)u(t-2)$
$- (3-t)u(t-3)$
$= (t-1)u(t-1) - 2(t-2)u(t-2)$
$+ (t-3)u(t-3)$

(iii) $\mathcal{L}\{f(t)\} = e^{-s}\mathcal{L}\{t+1-1\} - 2\,e^{-2s}\mathcal{L}\{t+2-23\} +$
$e^{-3s}\mathcal{L}\{t+3-3\}$

$$= e^{-s}\frac{1}{s^2} - 2\,e^{-2s}\frac{1}{s^2} + e^{-3s}\frac{1}{s^2}$$
$$= \frac{1}{s^2}(e^{-s} - 2\,e^{-2s} + e^{-3s}).$$

Example 1.23

Given $f(t) = \begin{cases} 9, & 0 < t < 2 \\ -6, & 2 \le t < 6 \\ 25, & 6 \le t < 9 \\ 7, & t > 9 \end{cases}$,

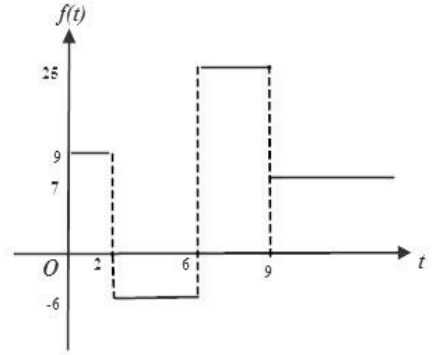
(i) sketch $f(t)$,

(ii) convert $f(t)$to unit step function and,

(iii) find the Laplace transformation of$f(t)$.

Solution:

(i)



(ii) $f(t) = 9[u(t-0) - u(t-2)] + (-6)[u(t-2) - u(t-6)] + 25[u(t-6) - u(t-9)] + 7[u(t-9)]$
$= 9u(t-0) - 15u(t-2) + 31u(t-6) - 18u(t-9)$

(iii) $\mathcal{L}\{f(t)\} = \mathcal{L}\{9u(t-0)\} - \mathcal{L}\{15u(t-2)\} + \mathcal{L}\{31u(t-6)\} - \mathcal{L}\{18u(t-9)\}$
$= 9e^{0s}\frac{1}{s} - 15e^{-2s}\frac{1}{s} + 31e^{-6s}\frac{1}{s} - 18e^{-9s}\frac{1}{s}$
$= (9 - 15e^{-2s} + 31e^{-6s} - 18e^{-9s})\frac{1}{s}.$

Short Impulses: Dirac's Delta Function

An airplane making a "hard" landing, a mechanical system being hit by a hammer-blow, a ship being hit by a single high wave, a tennis ball being hit by a racket, and many other similar examples appear in everyday life. They are phenomena of an impulsive nature where actions of forces mechanical, electrical, etc. are applied over short intervals of time.

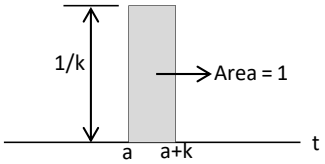
We can model such phenomena and problems by "Dirac's delta function," and solve them very effectively by the Laplace transform.

To model situations of that type, we consider the function

$$f_k(t-a) = \begin{cases} 1/k & if\ a \le t \le a+k \\ 0 & otherwise \end{cases}$$

(and later its limit as $k \to 0$). This function represents, for instance, a force of magnitude $\frac{1}{k}$ acting from $t = a$ to $t = a + k$, where k is positive and small. In mechanics, the integral of a force acting over a time interval $a \le t \le a + k$ is called the **impulse of the force**; similarly for electromotive forces $E(t)$ acting on circuits. Since the blue rectangle in the figure below has area 1, the impulse of f_k in (1) is

$$I_k = \int_0^\infty f_k(t-a)\,dt = \int_a^{a+k} \frac{1}{k}\,dt = 1$$



The function $f_k(t-a)$

To find out what will happen if k becomes smaller and smaller, we take the limit of f_k

as $k \to 0$ ($k > 0$). This limit is denoted by $\delta(t-a)$, that is,

$$\delta(t-a) = \lim_{k\to 0} f_k(t-a)$$

$\delta(t-a)$ is called the **Dirac delta function** or the **unit impulse function**.

$\delta(t-a)$ is not a function in the ordinary sense as used in calculus, but a so-called generalized function. To see this, we note that the impulse I_k of f_k is 1, by taking the limit as $k \to 0$ we obtain

$$\delta(t-a) = \begin{cases} \infty \text{ if } t = a \\ 0 \text{ otherwise} \end{cases}$$

and

$$\int_0^\infty \delta(t-a)\ dt = 1$$

but from calculus we know that a function which is everywhere 0 except at a single point just has the integral equal to 0. Nevertheless, in impulse problems, it is convenient to operate on $\delta(t-a)$ as though it were an ordinary function. For a continuous function $g(t)$ one uses the property often called the **sifting property** of $\delta(t-a)$, not to be confused with shifting

$$\int_0^\infty g(t)\delta(t-a)\,dt = g(a)$$

which is plausible.

To obtain the Laplace transform of $\delta(t\ -\ a)$, we write

$$f_k(t-a) = \frac{1}{k}\big[u(t-a) - u\big(t-(a+k)\big)\big]$$

and take the transform see (2)

$$\mathcal{L}\{f_k(t-a)\} = \frac{1}{ks}[\,e^{-as} - e^{-(a+k)s}] = e^{-as}\ \frac{1-e^{-ks}}{ks}$$

We now take the limit as $k \to 0$. By L'Hôpital's rule the quotient on the right has limit 1 (differentiate the numerator and the denominator separately with respect to k, obtaining se^{-ks}

and s, respectively, and use $\frac{se^{-ks}}{s} \to 1$ as $k \to 0$). Hence the right side has the limit e^{-as}. This suggests defining the transformation of $\delta(t-a)$ by this limit, that is,

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}$$

Example 1.24

Find the Laplace transformation of $f(t) = \delta(t-1) + cos\,2t\,\delta(t-3)$.

Solution: $\mathcal{L}\{\delta(t-1) + \cos 2t\,\delta(t-3)\}$

$= \mathcal{L}\{\delta(t-1)\} + \mathcal{L}\{\cos 2t\,\delta(t-3)\}$

$= e^{-s} + e^{-3s}\cos 6 \quad [\mathcal{L}\{f(t)\,\delta(t-a)\} = e^{-as} f(a)]$

Problem Set 1.2

Unit step function

1. Sketch the following functions and find their Laplace transforms:

$$\mathcal{L}\{f(t)\,u(t-a)\} = e^{-as}\,\mathcal{L}\{f(t+a)\}.$$

a. $f(t) = t\ u(t-1)$, **Ans:** $F(s) = \frac{e^{-s}}{s} + \frac{e^{-s}}{s^2}$.

b. $f(t) = (t-1)\,u(t-1)$, **Ans:** $F(s) = \frac{e^{-s}}{s^2}$

c. $f(t) = (t-1)^2\,u(t-1)$, **Ans:** $F(s) = \frac{2\ e^{-s}}{s^3}$.

d. $f(t) = e^{-2t}\,u(t-3)$, **Ans**: $F(s) = \frac{e^{-6}\ e^{-3s}}{s+2}$

e. $f(t) = 4\cos t\ u(t-\pi)$, **Ans:** $F(s) = -\frac{4s\ e^{-\pi s}}{s^2+1}$.

2. Sketch the following functions. Write $f(t)$ in terms of unit step function and hence find their Laplace transforms:

a. $f(t) = \begin{cases} t\,; 0 < t < 1 \\ 2; \quad t > 1 \end{cases}$

Ans: $f(t) = t\,u(t) + (2 - t)\,u(t-1)$.

$F(s) = \frac{1}{s^2} + e^{-s}\left(\frac{1}{s} - \frac{1}{s^2}\right)$.

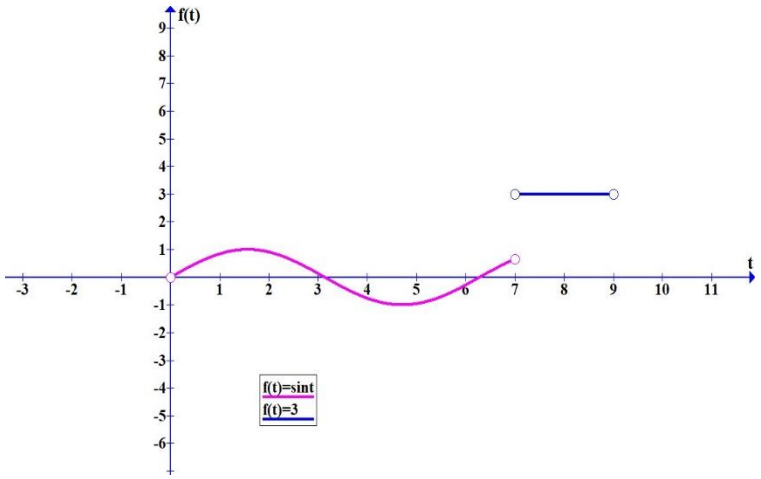
b. $f(t) = \begin{cases} t^2\,, \quad 0 \le t < 1 \\ t - 3, \quad t \ge 1' \end{cases}$

Ans: $f(t) = t^2\,u(t) + (t - 3 - t^2)\,u(t-1)$.

$F(s) = \frac{2}{s^3} - e^{-s}\left(\frac{3}{s} + \frac{1}{s^2} + \frac{2}{s^3}\right)$

3. Using the following graphs write $f(t)$ in terms of unit step function and hence find the Laplace transforms of $f(t)$:

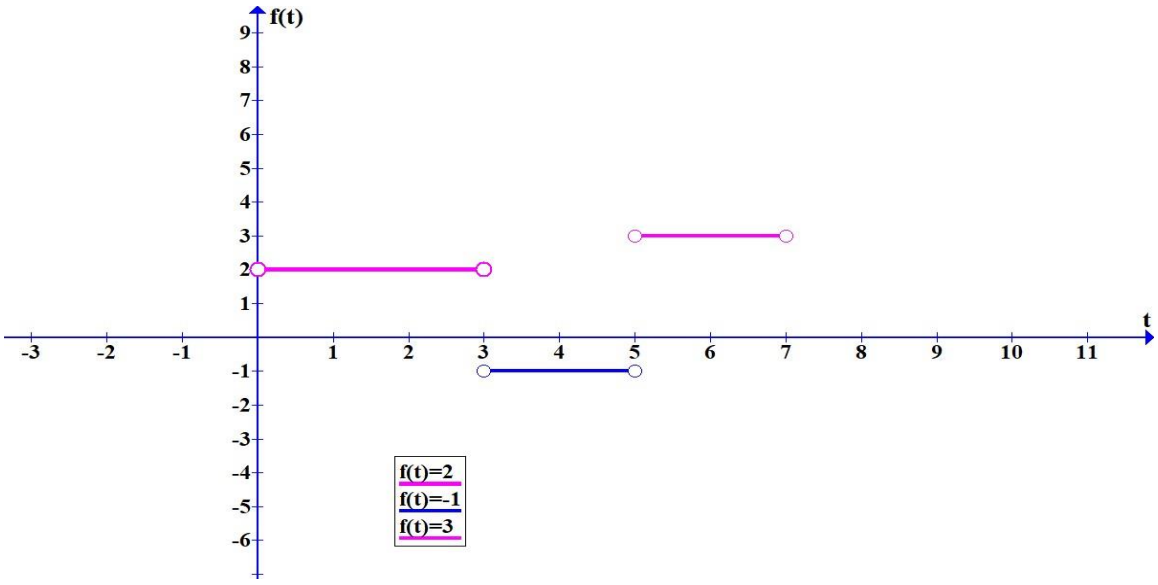
a.



Ans: $f(t) = \sin t\,\big(u(t) - u(t-7)\big) + 3\big(u(t-7) - u(t-9)\big)$.

$F(s) = e^{-7s}\left(\frac{3}{s} - \frac{\cos 7 + s \sin 7}{s^2+1}\right) - \frac{3\,e^{-9s}}{s} + \frac{1}{s^2+1}$.

b.



Ans: $f(t) = 2u(t) - 3u(t-3) + 4u(t-5) - 3u(t-7).$

$$F(s) = \frac{4\ e^{-5s}}{s} - \frac{3\ e^{-3s}}{s} - \frac{3\ e^{-7s}}{s} + \frac{2}{s}.$$

Dirac's delta (Unit impulse) function

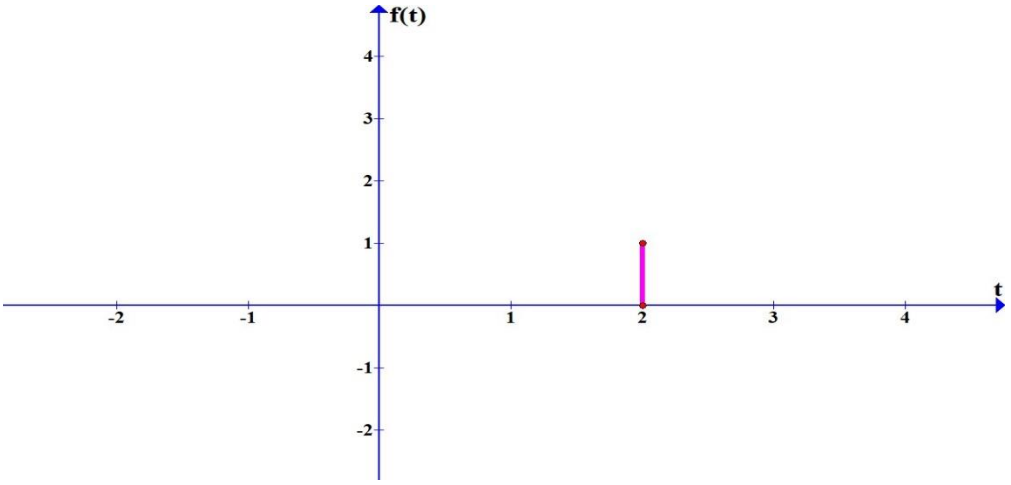
4. Sketch the following functions and find their Laplace transforms:

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}.$$

$$\mathcal{L}\{f(t)\ \delta(t-a)\} = e^{-as}\ f(a).$$

a. $f(t) = \delta(t-2),$

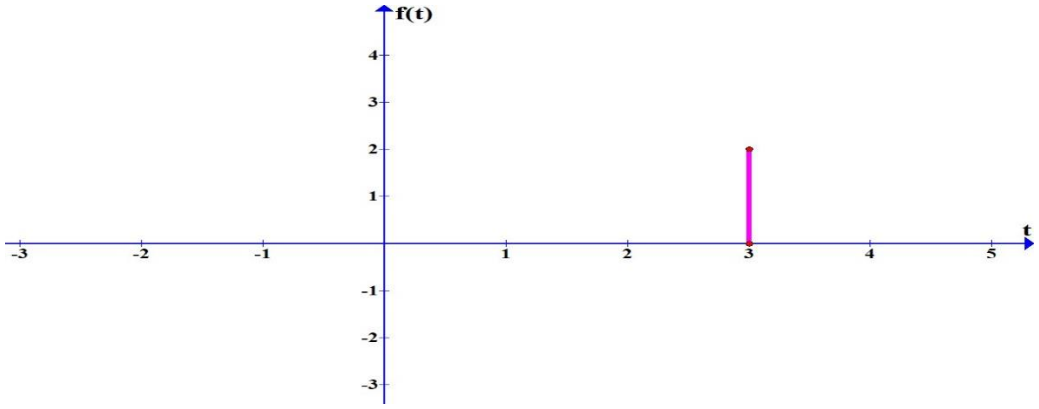
Ans:



$$F(s) = e^{-2s}.$$

b. $f(t) = 2\ \delta(t-3),$

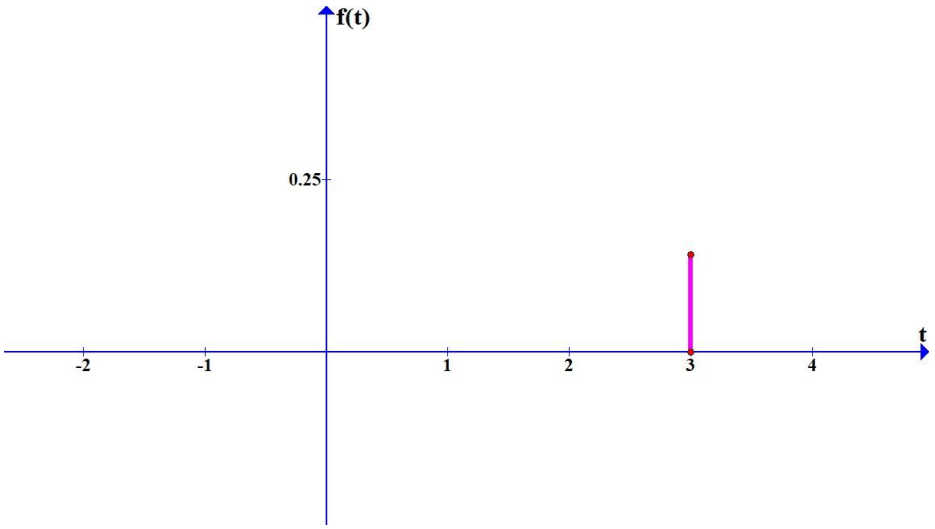
Ans:



$$F(s) = 2\ e^{-3s}.$$

c. $f(t) = \sin t\ \ \delta(t-3),$

Ans:



$$F(s) = \sin 3 \; e^{-3s}.$$

5. Problems from Book: (Ref: Advanced Engineering Mathematics – Erwin Kreyszig**) Convert $f(t)$ to Unit step function and find the Laplace transformations of $f(t)$:**

a. $f(t) = t\ (0 < t < 2)$

Ans: $\frac{1}{s^2} - e^{-2s}\left[\frac{1}{s^2} + \frac{2}{s}\right]$

b. $f(t) = t^2\left(t > \frac{3}{2}\right)$

Ans: $e^{-\frac{3}{2}s}\left[\frac{2}{s^3} + \frac{3}{s^2} + \frac{9}{4s}\right]$

c. $f(t) = t - 2\ (t > 2)$

Ans: $\frac{e^{-2s}}{s^2}$

d. $f(t) = sinh\, t\ (0 < t < 2)$

Ans: $\frac{1}{s^2-1} - e^{-2s}\left[\frac{\cosh 2}{s^2-1} + \frac{s\ \sinh 2}{s^2-1}\right]$

e. $f(t) = cos\, 4t\ (0 < t < \pi)$

Ans: $\frac{s}{s^2+16} - \frac{se^{-\pi s}}{s^2+16}$

f. $f(t) = sin\, t\left(\frac{\pi}{2} < t < \pi\right)$

Ans: $\frac{s\, e^{-\frac{\pi}{2}s}}{s^2+1} + \frac{e^{-\pi s}}{s^2+1}$

g. $f(t) = e^t\left(0 < t < \frac{\pi}{2}\right)$

Ans: $\frac{1}{s-1} - \frac{e^{(1-s)\frac{\pi}{2}}}{s-1}$

h. $f(t) = 16t^2 u\left(t - \frac{1}{4}\right)$

Ans: $16e^{-\frac{1}{4}s}\left[\frac{2}{s^3} + \frac{1}{2s^2} + \frac{1}{16s}\right]$

i. $f(t) = sin\, \pi t\ (2 < t < 4)$

Ans: $\frac{\pi e^{-2s}}{s^2+\pi^2} - \frac{\pi e^{-4s}}{s^2+\pi^2}$

j. $f(t) = e^{\frac{t}{2}} u(t-3)$

Ans: $e^{\left(\frac{3}{2}-3s\right)}\frac{2}{2s-1}$

k. $f(t) = e^{-\pi t}(2 < t < 4)$

Ans: $\left[e^{-2(s+\pi)} - e^{-4(s+\pi)}\right]\frac{1}{s+\pi}$

l. $f(t) = u(t-2\pi)\, sin\, t$

Ans: $\frac{e^{-2\pi s}}{s^2+1}$

m. $f(t) = t^2\ (1 < t < 2)$

Ans: $e^{-s}\left[\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right] - e^{-2s}\left[\frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s}\right]$